

WASP-52b

R-1693

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_210914 · created 2026-08-02T13:24:40+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459471.7295 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.178 +/- 0.016
Transit depth (Rp/Rs)^2	3.17 +/- 0.57 [%]
Orbital Inclination (inc)	89.75 +/- 1.15
Airmass coefficient 1 (a1)	0.957 +/- 0.024
Airmass coefficient 2 (a2)	0.029 +/- 0.014
Scatter in the residuals of the lightcurve fit is	2.5 %
Best Comparison Star	#3 - [312, 446]
Optimal Aperture	4.54
Optimal Annulus	10.98
Transit Duration (day)	0.0886 +/- 0.0016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.178 $\pm$ 0.016 vs 0.1646 (deviation 0.8 σ)
Duration fit vs published	0.0886 d vs 0.0758 d (deviation 7.62 σ)
Mid-time O-C	11.4 min
Depth SNR	10.6
Residual scatter	2.5 %

Light curve

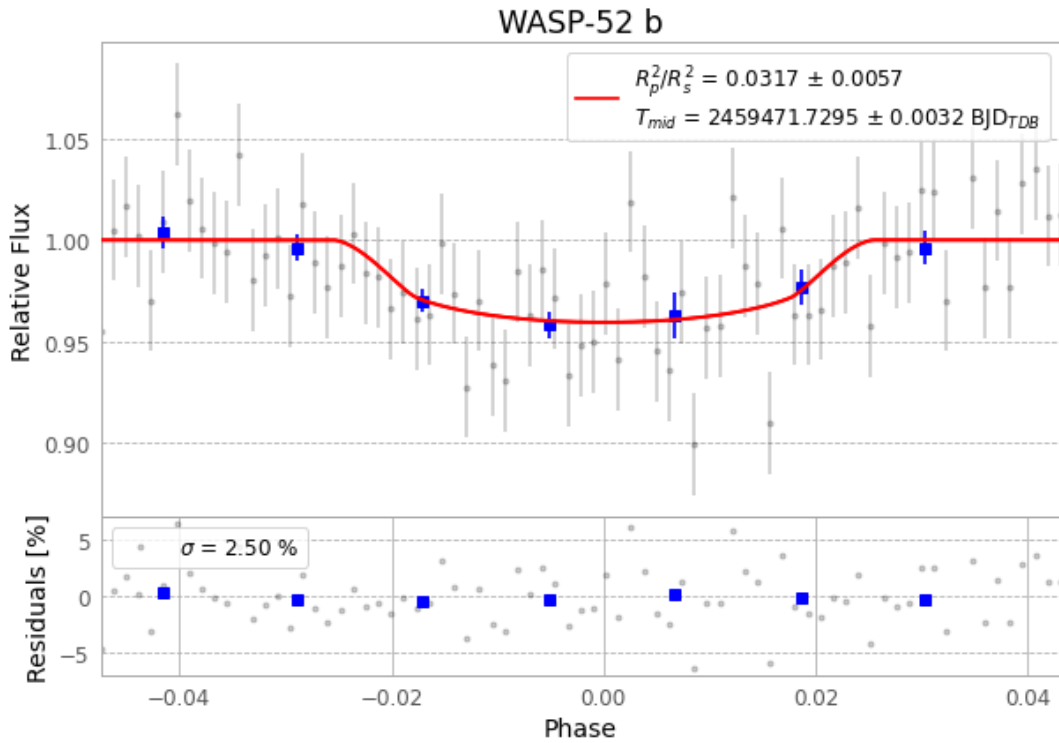


Figure 1 — FinalLightCurve_WASP-52 b_2021-09-13.png (SHA-256 66693c0d0252ab9f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 135], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 10ef87d403f29928...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52210914032112.FITS → WASP-52210914070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-210914.log (7b0b2183cc36...), FinalLightCurve_WASP-52 b_2021-09-13.png (66693c0d0252...), FinalLightCurve_WASP-52 b_2021-09-13.csv (7f75191104b1...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)